

Parag, A. S.¹, Marsellos, A.E.¹, Tsakiri, K.G.²

²Dept. Information Systems, Analytics, and Supply Chain Management, Rider University, NJ, USA.



Long Island has experienced subsidence since the last ice age when the large Laurentide ice sheet covered most of New York State and readjusted after it melted. This process of readjustment of the land is known as isostasy, the equilibrium position where a body of land should sit between the surface above and below. However, the region is not uplifting after the weight of the ice melted but rather continues to subside even after this event. This is due to human environmental interferences, particularly groundwater overpumping, causes uneven land deformation that heightens risks of landslides, sinkholes, and other geohazards (U.S. Geological Survey, n.d.). Central Long Island, with its many residential properties, has a high demand for groundwater, leading to the overpumping of aquifers in certain areas. This excessive withdrawal may contribute to gradual subsidence in parts of the region. In contrast, areas within New York City and Montauk Point exhibit significant subsidence rates despite lower residential groundwater extraction. This is likely due to reliance on upstate water sources, heavy infrastructure, and high population density. While Montauk Point, with its lower population density, experiences notable subsidence, possibly due to reduced aquifer pumping, which allows underlying geological formations to compact under their own weight. To capture these spatial variations in land motion, we analyzed continuous GPS records from multiple stations across differing time intervals. Rather than restricting our study to a common observation window, we considered each station's full operational timeline to assess overall trends. A GIS-interpolated map of average displacement rates reveals localized areas of significant subsidence, illuminating the combined effects of groundwater extraction, land use changes, and geological factors. This study underscores the need for strategic groundwater management and infrastructure planning to reduce further land subsidence and its associated hazards in this vulnerable coastal setting.

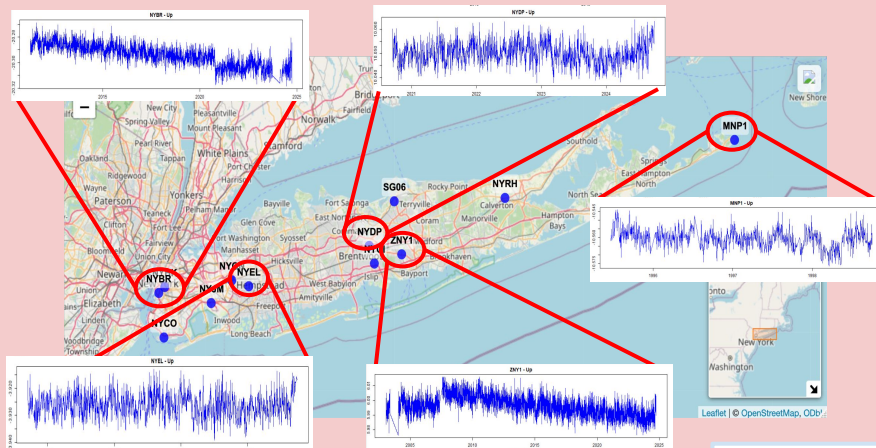


Figure 1: These geostatistical graphs display the up components of each station and their vertical movements within their various ranges. Some stations range from 1995-1998 and variations within the 2000s. The x-axis is time in years and the y-axis is the vertical displacement in meters (m).

- Long Island has been subsiding since the last ice age due to glacial isostatic adjustment.
- Glacial isostasy causes land to subside under ice weight and uplift when the ice melts.
- Unlike expected uplift, the region continues to subside, suggesting additional contributing factors.
- Groundwater overpumping may lead to uneven land deformation, increasing geohazard risks like landlides and sinkholes formed by underground erosion and sediment collapse.
- This study uses GPS data to detect ongoing vertical land movements beyond 2005–2020 research.
- GIS-interpolated maps highlight localized subsidence patterns, linking natural and human-induced factors.
- Long-term geostatistical analysis will reveal patterns of land movement and subsidence through vertical displacement rates.
- Findings will help understand isostasy's role and support efforts to mitigate environmental risks.

➤ Displays vertical movements of stations over time (1995-1998 & 2000s)

- Highest subsidence (~2 mm/yr) near NYC & eastern Long Island; central Long Island shows lower subsidence with occasional uplift.

- NYBR (NYC): Strong long-term subsidence.
- MNP1 (Eastern Long Island): Gradual downward trend but higher subsidence rates than central stations.
- NYEL, NYPD, ZNY1 (Central Stations): Lower subsidence rates with slight uplift periods.

- Up Component: Multiple peaks, indicating periodic oscillations in vertical displacement. Clustering of peaks suggests seasonal and sub-annual cycles.
- North Component: Low but visible long-term periodic movements in the northward direction.
- East Component: Shows long-wavelength trends, potentially linked to tectonic or subsidence factors.

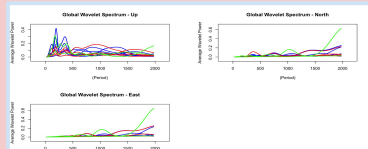


Figure 2. These graphs display the up, north, and east components of each station and their oscillations (various colored lines) within their different ranges. Some stations range from 1995–1998 and variations within the 2000s. The x-axis is the period (in days) and the y-axis is the vertical displacement in meters (m).

Figure 3 This GIS map displays the vertical displacement rates of each station and reveals potential areas of subsidence or uplift. The light red and white colored areas reveal areas of subsidence, the dark red (colored areas) reveal areas of uplift, and the blue colored areas reveal areas of high subsidence. Interactive map created in RStudio.

- GPS displacement data from 12 Long Island stations were retrieved using R from the Nevada Geodetic Laboratory. (Fig. 1)
- Data processing and cleaning used R packages.
- Custom function (GPS_dwd) structured data into northward, eastward, and upward displacement components.
- Missing values were interpolated, and stations with excessive data issues (MOB6, QJNS) were excluded.
- Outliers were removed using 2-sigma normalization (values beyond 3 standard deviations).
- Noise reduction applied using Kolmogorov-Zurbenko (KZ) filtering (365-day window, 3 replications).
- Wavelet Splinelet (WaveletCom) detected periodic signals in Up, North, and East displacement components. (Fig. 2)
- Linear regression (lm function) calculated vertical displacement rates from onset to end of the study period.
- Geospatial interpolation (IDW method) mapped the great up and upback rates.
- Near visualization (leaflet package) created an interactive map showing GPS station locations and displacement rates. (Fig. 3)

I would like to thank Dr. Antonius E. Marsellis and the Geology, Environment and Sustainability department at Hofstra University for providing me with the necessary tools to conduct this research. I would also like to thank Dr. Tsakiri for providing me with insights on KZ filtering system.

Parag, A.: Literature, Editing, RStudio Coding, Writing- Abstract, Introduction, Methods, Results-Figures, Discussion, Conclusion; Marsellos, A.E.: Revision, Editing, RStudio Coding; Taskiri, K.G.: KZ script insights

- NYC has the highest subsidence due to infrastructure weight and limited groundwater extraction.
- Central Long Island shows slight uplift, likely from groundwater overpumping.
- Eastern Long Island subsides despite low population, possibly due to retained groundwater weight.
- Subsidence may be influenced by groundwater and human activity, not tectonic factors.
- GPS data analysis confirms sinking rates, averaging 1.6 mm/year, with some areas exceeding 3 cm/decade.
- KZ filtering (365-day) removes noise, preserving long-term displacement trends.
- Subsidence varies across regions, with NYC and Montauk Point sinking more than central Long Island.
- Weier spectrum analysis identifies periodic land displacement possibly influenced by groundwater, temperature, and subsidence.

[illegible]